

FILE EXPLORER APPLICATION (CAPSTONE_1)

```
Open file_explorer.cpp Save
1 #include <iostream>
2 #include <filesystem>
3 #include <fstream>
4 #include <string>
5 #include <unistd.h>
6 #include <sys/stat.h>
7 #include <string>
8 #include <cstdlib>
9
10 namespace fs = std::filesystem;
11
12 void listFiles(const fs::path& path) {
13     std::cout << "Listing files in: " << path << "\n";
14     for (const auto& entry : fs::directory_iterator(path)) {
15         std::cout << (fs::is_directory(entry) ? "[DIR] " : " ");
16         std::cout << entry.path().filename().string() << "\n";
17     }
18 }
19
20 void changeDirectory(fs::path& currentPath) {
21     std::string dirName;
22     std::cout << "Enter directory name to navigate (.. for up): ";
23     std::cin >> dirName;
24     fs::path newPath = currentPath / dirName;
25     if (fs::exists(newPath) && fs::is_directory(newPath)) {
26         currentPath = fs::canonical(newPath);
27     } else {
28         std::cout << "Directory not found.\n";
29     }
30 }
31
32 void createFile(const fs::path& currentPath) {
33     std::string fileName;
34     std::cout << "Enter file name to create: ";
35     std::cin >> fileName;
36     std::ofstream file(currentPath / fileName);
37     if (file) {
38         std::cout << "File created successfully.\n";
39     } else {
40         std::cout << "Error creating file.\n";
41     }
42 }
43
44 void deleteFile(const fs::path& currentPath) {
45     std::string fileName;
46     std::cout << "Enter file/directory name to delete: ";
47     std::cin >> fileName;
48     fs::path target = currentPath / fileName;
49     try {
50         fs::remove_all(target);
51         std::cout << "Deleted successfully.\n";
52     } catch (...) {
53         std::cout << "Error deleting.\n";
54     }
55 }
56
```

```
This message is shown once a day. To disable it please create the
/home/cosmic/.hushlogin file.
cosmic@COSMIC: /mnt/c/Users/mohan$ gedit file_explorer.cpp
cosmic@COSMIC: /mnt/c/Users/mohan$ g++ -std=c++17 file_explorer.cpp -o file_explorer
cosmic@COSMIC: /mnt/c/Users/mohan$ ./file_explorer
```

Current Path: "/mnt/c/Users/mohan"

--- File Explorer Menu ---

1. List files
 2. Change directory
 3. Create file
 4. Delete file/directory
 5. Copy file
 6. Move/Rename file
 7. Search file
 8. Change file permissions
 9. Exit
- Choose an option: 1

Listing files in: "/mnt/c/Users/mohan"

```
[DIR] .arduinoIDE
[DIR] .aws
[DIR] .azure
[DIR] .bash_history
[DIR] .docker
[DIR] .gitconfig
[DIR] .lessht
[DIR] .minttyrc
[DIR] .vscode
[DIR] AppData
[DIR] Application Data
[DIR] Contacts
[DIR] Cookies
[DIR] Documents
[DIR] Downloads
[DIR] Favorites
[DIR] file_explorer
[DIR] file_explorer.cpp
[DIR] Links
[DIR] Local Settings
[DIR] Music
[DIR] My Documents
```

```
Current Path: "/mnt/c/Users/mohan"

--- File Explorer Menu ---
1. List files
2. Change directory
3. Create file
4. Delete file/directory
5. Copy file
6. Move/Rename file
7. Search file
8. Change file permissions
9. Exit
Choose an option: 2
Enter directory name to navigate (.. for up): my-website

Current Path: "/mnt/c/Users/mohan/my-website"

--- File Explorer Menu ---
1. List files
2. Change directory
3. Create file
4. Delete file/directory
5. Copy file
6. Move/Rename file
7. Search file
8. Change file permissions
9. Exit
Choose an option: 3
Enter file name to create: 2141013232_LOS_ASSIGNMENT_1
File created successfully.
```

```
--- File Explorer Menu ---
1. List files
2. Change directory
3. Create file
4. Delete file/directory
5. Copy file
6. Move/Rename file
7. Search file
8. Change file permissions
9. Exit
Choose an option: 4
Enter file/directory name to delete: 2141013232_LOS_ASSIGNMENT_1
Deleted successfully.

Current Path: "/mnt/c/Users/mohan/my-website"

--- File Explorer Menu ---
1. List files
2. Change directory
3. Create file
4. Delete file/directory
5. Copy file
6. Move/Rename file
7. Search file
8. Change file permissions
9. Exit
Choose an option: 5
Enter source file name: my-website
Enter destination file name: my-bootstrap-website
Error copying file.

Current Path: "/mnt/c/Users/mohan/my-website"

--- File Explorer Menu ---
1. List files
2. Change directory
3. Create file
4. Delete file/directory
5. Copy file
6. Move/Rename file
7. Search file
8. Change file permissions
9. Exit
Choose an option: 6
```

```
Current Path: "/mnt/c/Users/mohan/my-website"

--- File Explorer Menu ---
1. List files
2. Change directory
3. Create file
4. Delete file/directory
5. Copy file
6. Move/Rename file
7. Search file
8. Change file permissions
9. Exit
Choose an option: 9
Exiting...
cosmic@COSMIC:/mnt/c/Users/mohan$ |
```